

Scalability & Future Plan

Date	2 July, 2026
Team ID	
Project Name	Rising Waters – Flood Prediction System using AIML
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current Rising Waters – Flood Prediction System can be scaled to support larger datasets, more concurrent users, and additional intelligent features in the future. It also presents the team's roadmap for enhancing the application beyond its current implementation.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address(High / Medium / Low)
1	Uses historical weather data only (no live weather data integration)	Predictions cannot reflect real-time weather conditions.	High
2	Single-server deployment with no load balancing	Performance may decrease with a large number of simultaneous users.	Medium
3	Limited weather parameters used for prediction.	Prediction accuracy may be affected in complex weather scenarios.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports around 100 concurrent users on a single instance	Deploy on cloud (AWS/Azure) with auto-scaling and load balancers to support 10,000+ concurrent users
2	Data Storage	Weather dataset stored locally.	Migrate to scalable cloud storage (Amazon RDS / Azure SQL) with partitioning and backups
3	Performance	Synchronous request processing causing delays under load	Introduce asynchronous processing and a Redis caching layer to reduce latency
4	Security	Manual weather data entry through the web application.	Integrate real-time weather APIs (IMD/OpenWeatherMap) for automatic data collection and prediction.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target timeline	Expected Impact
Phase-2	Integrate real-time weather APIs for automatic weather data collection.	04/07/2026	More accurate and real-time flood predictions.
Phase-3	Add GIS maps, location-based flood alerts, and SMS/Email notifications.	04/07/2026	Faster disaster response and improved public safety.
Phase-4	Develop a mobile application and enhance prediction using Deep Learning models (LSTM/ANN).	04/07/2026	Better accessibility, improved prediction accuracy, and wider user adoption.